



UNIVERSITY OF GRANADA

Faculty of Science

Monte Carlo Simulation of Beta Radiation Attenuation for Industrial Paper Grammage Control

Author:

Marco Fumagalli

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Contents

Beta Radiation Analysis	1
1 Theoretical Background	2
2 Study Objective	2
2.1 Absorber Material Properties	2
2.2 Simulation Geometry	3
2.3 Data Analysis	3
3 Krypton-85	4
4 Thallium-204	4
5 Strontium-90	5
6 Why Beta Radiation?	6
6.1 Comparison with Gamma Rays	6
6.2 Limitations of Beta Sensors	8
Discussion of Results	8
7 Comparison of linear attenuation coefficients	9
8 Answers to the Proposed Questions	9

Beta Radiation Analysis

1. Theoretical Background

Beta particles are fast-moving electrons emitted by radioactive nuclei. Unlike heavy charged particles, beta particles are emitted with a continuous energy spectrum ranging from near zero up to a maximum endpoint energy (E_{max}), characteristic of the specific isotope. As these electrons travel through an absorber material, they lose energy primarily through inelastic collisions with atomic electrons (causing ionization and excitation) and undergo significant angular deflections due to elastic scattering with atomic nuclei.

Because of the continuous nature of their emission spectrum and the highly tortuous paths they take through matter (multiple Coulomb scattering), the macroscopic transmission of a beta particle beam through a thin absorber empirically follows an exponential attenuation law, similar to that of electromagnetic radiation:

$$I(x) = I_0 e^{-\mu x} \quad (1)$$

where $I(x)$ is the transmitted particle intensity, I_0 is the initial intensity, x is the thickness of the absorber, and μ [cm⁻¹] is the linear attenuation coefficient.

For industrial applications such as paper manufacturing, it is often more practical to express this relationship using the mass attenuation coefficient (μ/ρ), where ρ is the density of the material. The attenuation coefficient is heavily dependent on the E_{max} of the source: isotopes with a higher E_{max} exhibit a lower μ , resulting in greater penetrating power. Selecting the correct isotope is crucial; the attenuation must be perfectly balanced so that the radiation is sensitive to microscopic variations in paper thickness without being completely absorbed or fully transmitted.

2. Study Objective

To evaluate the theoretical principles of beta attenuation, this study utilizes the PENELOPE Monte Carlo code to simulate the transmission of beta radiation from three different sources (⁸⁵Kr, ²⁰⁴Tl, and ⁹⁰Sr) through varying thicknesses of an absorber. By simulating a range from 1 to 15 paper sheets (assuming an increment of 0.1 mm per sheet, up to a total thickness of 1.5 mm), we aim to reproduce the exponential attenuation curves, extract the exact linear attenuation coefficient μ for each isotope, and highlight how the source's energy spectrum dictates the sensitivity of the quality control process.

2.1. Absorber Material Properties

For these simulations, Cellulose Acetate Butyrate was selected as the reference absorber material, accurately modeled within the PENELOPE material database. The key physical and

atomic properties extracted from the database and implemented in the simulation are summarized in Table 1.

Property	Value	Description
Mass Density (ρ)	1.20 g/cm ³	Standard reference value for dense cellulosic materials.
Mean Excitation Energy (I)	74.6 eV	The average energy required to excite or ionize the material's atoms.

Table 1: Physical and atomic properties of the Cellulose Acetate Butyrate absorber.

2.2. Simulation Geometry

The geometry of the experimental setup was characterized by the following key parameters:

- **Target Absorber (CYLIND):** Modeled as a wide, disk-like cylinder with a massive lateral radius of 10.0 cm.¹ In contrast, the vertical thickness of this disk (defined along the Z-axis, starting from $Z = 0.0$ cm) was the independent variable, iterating from 0.1 mm to 1.5 mm.
- **Beta Source (SKPAR 1):** Positioned at $(0, 0, -0.1)$ cm, creating a precise 1 mm gap below the absorber's bottom surface.
- **Isotropic Emission (SCONE 0 0 180):** The source was configured to emit particles isotropically in all directions (a full sphere). Consequently, exactly 50% of the primary particles are initially directed upwards to strike the sample, while the remaining half are emitted downwards and immediately lost.

2.3. Data Analysis

To ensure the reliability of the extracted attenuation parameters, the simulation data for each isotope was processed and validated using two complementary mathematical approaches:

1. **Linear Scale Analysis:** The transmitted particle fraction is plotted on a standard linear scale against the absorber thickness. An exponential regression ($I = I_0 e^{-\mu x}$) is applied directly to the data points, yielding a direct estimation of μ .
2. **Semi-logarithmic Scale Analysis:** The data is transformed and displayed on a semi-logarithmic scale. By applying a linear fit to the natural logarithm of the transmitted fraction ($\ln(I/I_0) = -\mu x + q$), the slope of the resulting line provides a secondary, robust validation of the attenuation coefficient.

Applying this dual analysis to the simulation data allows for a robust extraction of the attenuation coefficients (μ and μ/ρ), as detailed in the following sections starting with Krypton-85.

¹This large horizontal extension ensures the sample acts as an infinite plane relative to the beam, completely preventing scattered particles from escaping sideways (edge effects).

3. Krypton-85

Krypton-85 is a radioactive isotope of a noble gas that decays into stable Rubidium-85 via beta-minus (β^-) emission. The primary decay process is described by the following nuclear equation:



where β^- is the emitted electron and $\bar{\nu}_e$ represents the electron antineutrino. This transition accounts for the vast majority of decays and is characterized by a continuous energy spectrum with a maximum endpoint energy (E_{max}) of 687 keV.

The detailed ${}^{85}\text{Kr}$ spectrum used for the simulation was obtained from the LNH database². To ensure high statistical precision, the cellulose film was bombarded with 10^7 primary electrons for each thickness increment.

The results of the simulation and the subsequent data analysis are shown in Figure 1.

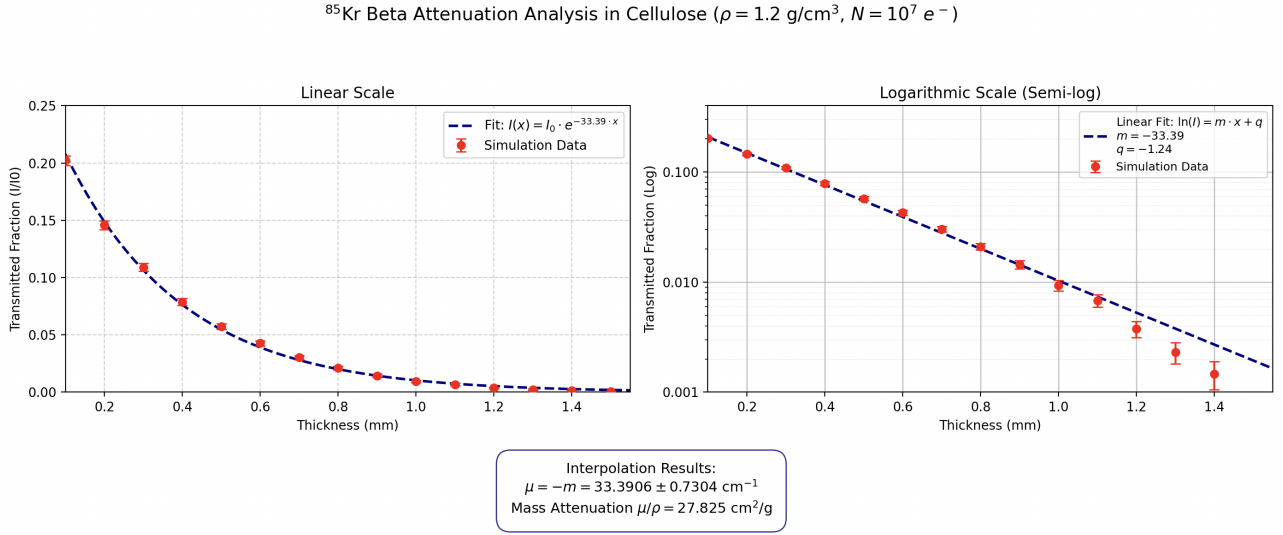


Figure 1: Beta attenuation analysis for ${}^{85}\text{Kr}$. Left: Exponential fit on a linear scale. Right: Linear fit on a semi-logarithmic scale.

From the fits, a linear attenuation coefficient of

$$\mu \approx 33.00 \text{ cm}^{-1} \quad (3)$$

was determined. Given the cellulose density used in the simulation ($\rho = 1.2 \text{ g/cm}^3$), the mass attenuation coefficient is calculated as:

$$\frac{\mu}{\rho} \approx 27.50 \text{ cm}^2/\text{g} \quad (4)$$

4. Thallium-204

Thallium-204 is a widely used solid beta source that primarily undergoes beta-minus decay to become stable Lead-204. This predominant decay branch, which accounts for over 97% of its disintegrations, is represented by the following nuclear equation:



²<http://www.lnhb.fr/nuclear-data/nuclear-data-table/>

where the (β^-) particle has a maximum energy of 763 keV.

Applying the established simulation procedure to the ^{204}Tl spectrum, we analysed its transmission through the cellulose layers using 10^7 primary electrons. As shown in Figure 2, the data confirms the expected exponential behaviour.

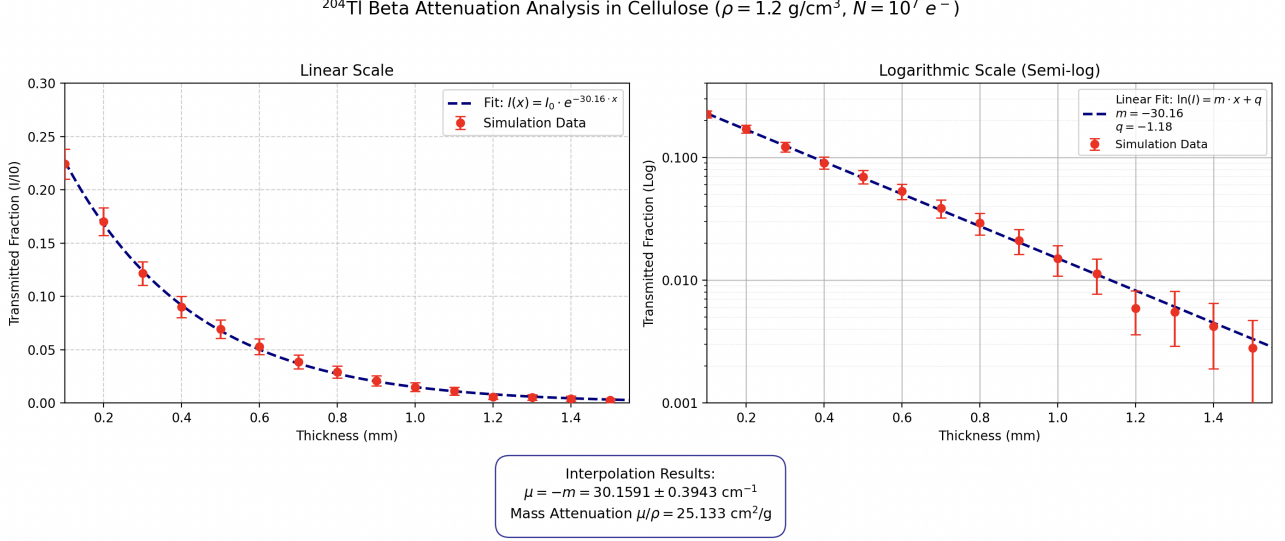


Figure 2: Attenuation analysis for ^{204}Tl . The higher endpoint energy compared to ^{85}Kr results in a visibly less steep slope.

As outlined in the methodology, the fits on both linear and semi-logarithmic scales yielded a consistent linear attenuation coefficient of:

$$\mu \approx 30.00 \text{ cm}^{-1} \quad (6)$$

With a cellulose density of $\rho = 1.2 \text{ g/cm}^3$, the resulting mass attenuation coefficient is calculated as:

$$\frac{\mu}{\rho} \approx 25.00 \text{ cm}^2/\text{g} \quad (7)$$

5. Strontium-90

Strontium-90 is a radioactive metal characterized by a half-life of approximately 28.8 years. It undergoes beta-minus decay to become Yttrium-90, as described by the following nuclear equation:



Following the methodology established for the previous isotopes, we simulated the emission of 10^7 primary electrons from the pure ^{90}Sr source through the cellulose layers.

The simulation results yield a linear attenuation coefficient of

$$\mu \approx 47.60 \text{ cm}^{-1}, \quad (9)$$

as shown in Figure 3. This value is significantly higher than those calculated for Krypton-85 and Thallium-204, confirming that the ^{90}Sr beta spectrum is much more susceptible to absorption within the cellulose fibers due to its lower average energy.

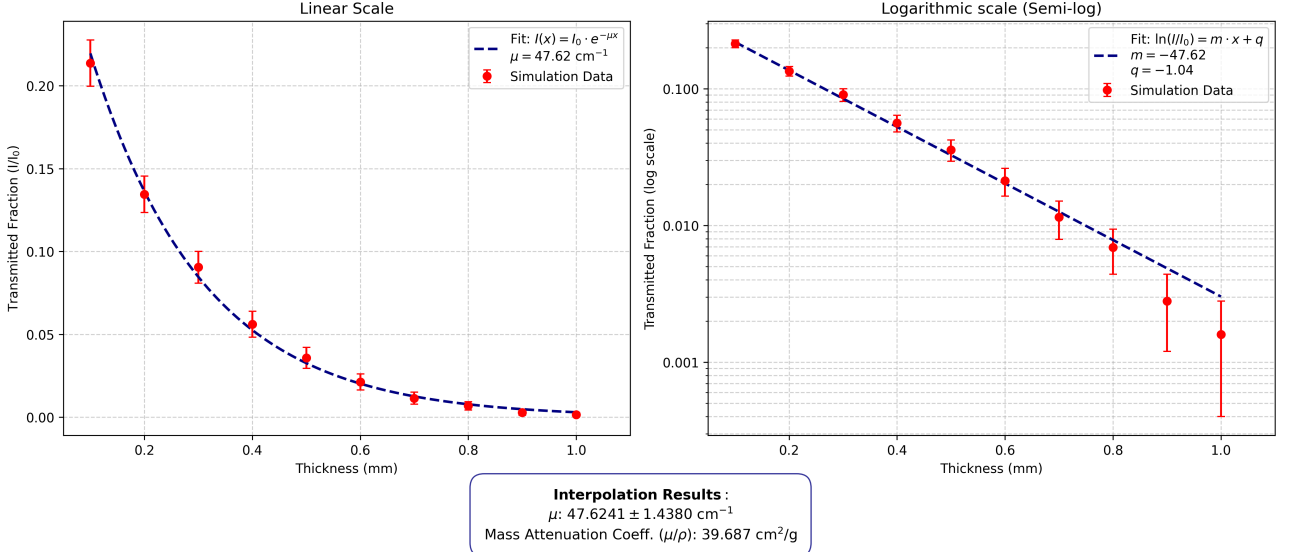


Figure 3: Beta attenuation analysis for ⁹⁰Sr. The steep slope in both plots reflects the rapid absorption of the low-energy electrons emitted by this nuclide.

For the ⁹⁰Sr analysis, the data interpolation was deliberately restricted to a maximum thickness of 1.0 mm. Beyond this depth, the low-energy beta particles are virtually entirely absorbed ($I(x) \rightarrow 0$).

The resulting mass attenuation coefficient is calculated as:

$$\frac{\mu}{\rho} \approx 40.50 \text{ cm}^2/\text{g} \quad (10)$$

Having established the attenuation behaviour of these three distinct beta emitters, it is clear how their continuous energy spectra are perfectly suited for measuring thin layers of cellulose. However, to fully appreciate the engineering choice behind these industrial sensors, a fundamental question remains:

Why exclusively use beta particles?

The following section addresses this by directly comparing the beta attenuation profile with that of highly penetrating gamma radiation.

6. Why Beta Radiation?

6.1. Comparison with Gamma Rays

To fully understand why beta radiation is the industry standard, it is instructive to propose a comparative model. In this theoretical exercise, I simulated a beam of photons (gamma rays) colliding with the cellulose target, assigning them energies equivalent to the maximum beta energies of ⁹⁰Sr, ⁸⁵Kr, and ²⁰⁴Tl (546 keV, 687 keV, and 763 keV, respectively).

Unlike beta particles, high-energy photons interact very weakly with low-density, low-Z materials like cellulose. Consequently, they possess a **very low** linear attenuation coefficient μ . This physical reality becomes visually evident when analysing the transmission graphs: the linear scale plot and the semi-logarithmic scale plot appear almost identical.

The data interpolation is presented in Figure 4: here we can clearly see that the attenuation profiles for the three energy equivalents are essentially identical. Throughout the simulated thickness range, the transmitted fraction remains constantly near 0.5³.

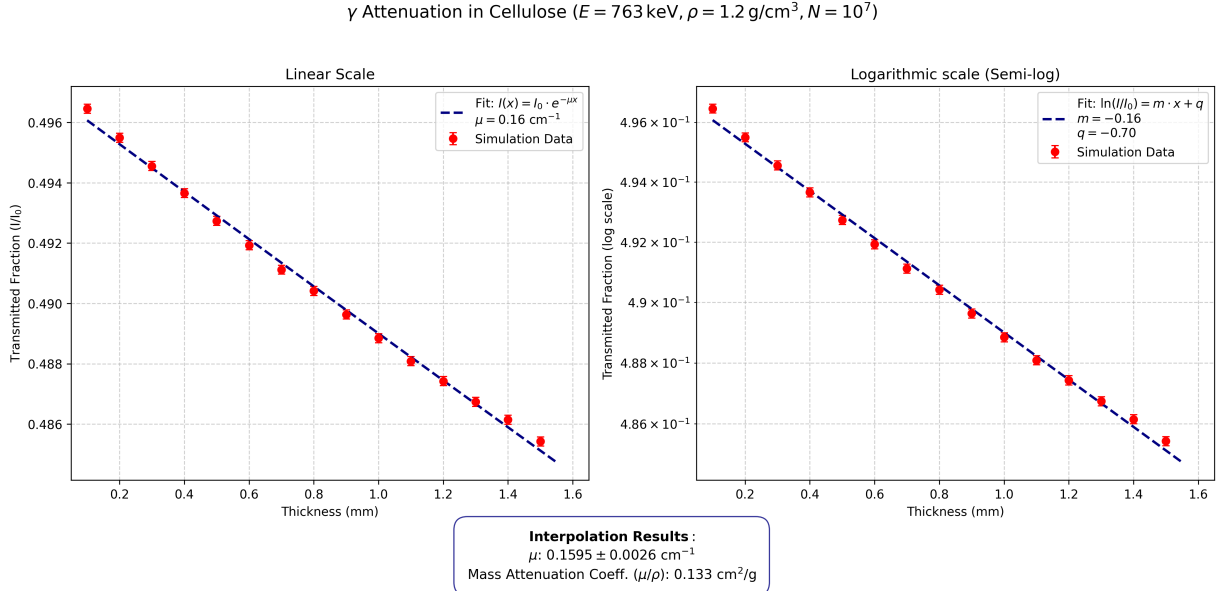


Figure 4: Attenuation analysis for Gamma radiation (763 keV). The practically flat slope demonstrates the near-total transmission of the photons, making the sensor blind to thickness variations.

Mathematically, since the attenuation coefficient μ is exceptionally small, the product $\mu x \ll 1$ for the paper thicknesses involved. Therefore, the exponential function can be approximated at the first order as:

$$\frac{I(x)}{I_0} = e^{-\mu x} \approx 1 - \mu x \quad (11)$$

As shown by this expansion, the exponential curve degrades into a simple straight line with a very shallow slope $-\mu$. Although the simulation was performed for all three energy equivalents, the resulting transmission datasets are practically overlapping and visually indistinguishable. For this reason, Figure 4 displays only the results for the highest energy (^{204}Tl , 763 keV), serving as the representative upper bound of our energy range.

Indeed, it is now clear that utilizing gamma radiation for grammage control is a poor engineering choice for two fundamental reasons:

1. **Lack of Sensitivity:** The vast majority of the photons pass straight through the paper without interacting. Because the signal is not attenuated, the sensor would be entirely blind to microscopic thickness variations, rendering quality control impossible.
2. **Industrial Safety:** Gamma rays are highly penetrating and require massive, heavy lead shielding to protect factory personnel. In contrast, beta radiation is easily absorbed by thin layers of plastic or light metal, making sensor encapsulation significantly safer, cheaper, and more manageable.

³This baseline value of 50% occurs because the simulated source was configured to emit photons isotropically in all directions (SCONE 0 0 180); consequently, only half of the primary particles are initially directed upwards toward the cellulose layer and the detector, passing through it with virtually zero attenuation.

Having established beta radiation as the optimal choice for paper manufacturing, a critical question arises:

Are beta sensors a universal solution for all industrial materials?

6.2. Limitations of Beta Sensors

While beta radiation is perfectly tailored for thin, low-density materials like cellulose, its physical properties become a severe limitation when applied to heavy, dense absorbers. To empirically establish the operative limits of beta sensors, a complementary simulation was performed using Lead ($Z = 82$, $\rho = 11.35 \text{ g/cm}^3$) as the target material.

For this test, only the ^{204}Tl beta spectrum was simulated. This choice was deliberate: possessing the highest endpoint energy ($E_{\text{max}} = 763 \text{ keV}$) among the studied isotopes, it represents the most penetrating beta radiation available in our setup. If the most energetic beta beam is rapidly stopped by the absorber, it logically follows that the lower-energy emissions from ^{85}Kr and ^{90}Sr would be attenuated even more abruptly, rendering their simulation redundant. To accurately capture this rapid energy depletion, the geometric parameters of the simulation had to be drastically adjusted: while the cellulose sheets were evaluated over a thickness range of 0.1 mm to 1.5 mm, the lead absorber required a much finer microscopic scale, ranging from merely 0.01 mm up to 0.15 mm.

As depicted in Figure 5, even within this extremely thin range, the transmitted fraction of beta particles drops abruptly, reaching absolute zero before 0.15 mm. The incident beam is completely absorbed almost instantly.

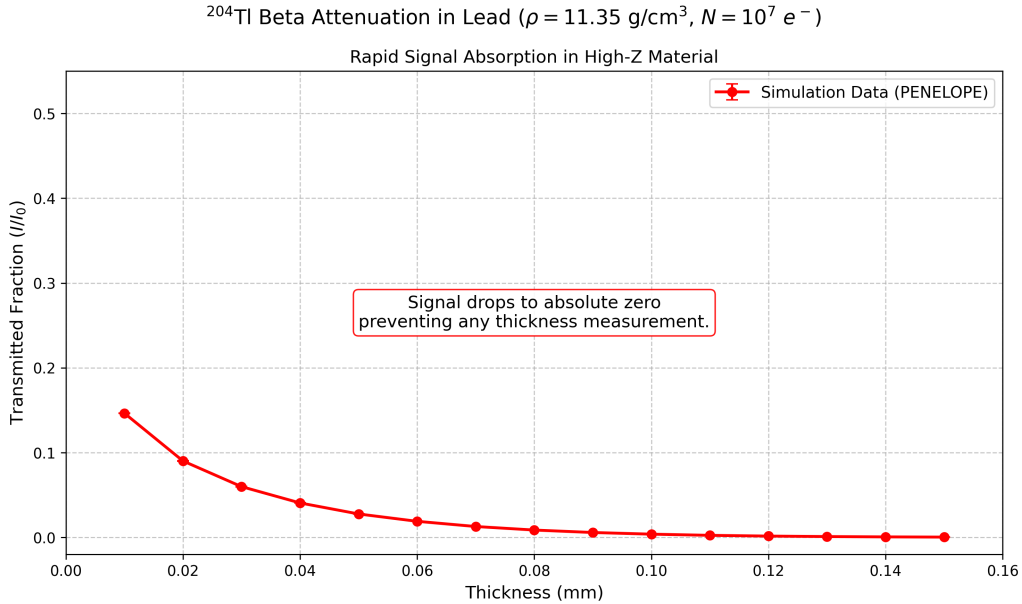


Figure 5: Beta attenuation of ^{204}Tl in Lead.

This dramatic signal loss dictates a fundamental rule in industrial quality control. For manufacturing processes involving thick or heavy materials, such as metal sheets in steel or lead rolling mills, beta radiation is entirely unsuitable, as the sensor would receive zero signal and be unable to detect any thickness variations. In these specific applications, the engineering paradigm must flip: the highly penetrating gamma rays, which failed in the paper industry due to lack of interaction, become the mandatory standard, as they possess the necessary penetrating power to provide a measurable and sensitive attenuation profile through dense metals.

Discussion of results

7. Comparison of linear attenuation coefficients

To summarize the findings of this study, the linear attenuation coefficients obtained for the three beta sources are compared. The following table illustrates the clear relationship between the maximum energy of the emission spectrum E_{max} and the resulting attenuation in cellulose.

Isotope	E_{max} (keV)	μ (cm ⁻¹)	μ/ρ (cm ² /g)	Penetrating Power
⁹⁰ Sr	546	47.60	40.50	Lowest
⁸⁵ Kr	687	33.00	27.50	Intermediate
²⁰⁴ Tl	763	30.00	25.00	Highest

Table 2: Comparison of experimental attenuation parameters for the studied isotopes.

8. Answers to the Proposed Questions

The experimental data presented in Table 2 confirms the fundamental principle of beta interaction with matter: the attenuation coefficient μ is inversely proportional to the maximum energy of the emission spectrum E_{max} . As E_{max} increases, the probability of interaction per unit path length decreases, resulting in more penetrating radiation. Based on these findings, we can conclusively address the guiding questions regarding the industrial application of these sources:

1. ¿Qué radiación proporciona dicha fuente radiactiva?

The sources analyzed in this study provide **beta-minus (β^-) radiation**. This radiation consists of high-speed electrons emitted from the nucleus with a continuous energy spectrum. Beta radiation is chosen because alpha particles would be completely absorbed by the air or the very first layer of paper, while gamma rays would pass through the thin cellulose sheets almost entirely without measurable attenuation.

2. ¿Qué ventajas proporciona el uso de esa fuente?

Taking ⁸⁵Kr as the primary example for paper manufacturing, its use provides three major advantages:

- **Optimal Sensitivity:** Its intermediate energy ($E_{max} = 687$ keV) yields an attenuation coefficient of ≈ 33 cm⁻¹, providing the perfect balance: the signal is neither fully absorbed nor fully transmitted by standard paper, allowing for highly precise grammage calculations.
- **Safety:** Being a noble gas, if the sensor window is compromised, the ⁸⁵Kr safely dissipates

into the atmosphere, preventing permanent radioactive contamination of the paper web and machinery.

- **Stability:** Its half-life of 10.76 years ensures long-term calibration stability.

3. ¿Cómo funciona el control del proceso de fabricación del papel haciendo uso del dispositivo descrito en la figura?

The process control operates on the physical principle of radiation transmission. A beta source is positioned on one side of the moving paper web, and a detector is aligned opposite to it. As the paper passes between them, it absorbs a fraction of the beta particles. The detector continuously measures the transmitted intensity $I(x)$. Using the exponential attenuation law

$$I(x) = I_0 e^{-\mu x},$$

the system calculates the material's thickness in real-time. This continuous measurement is then sent to a control system that automatically adjusts the pressure of the rollers to achieve and maintain the desired target grammage.

4. ¿Qué otras fuentes se usan en la industria del papel? ¿Qué razón es la fundamental para elegir una fuente u otra?

In addition to ^{85}Kr , other standard sources widely used in the paper industry include Strontium-90 (^{90}Sr) and Thallium-204 (^{204}Tl).

The fundamental reason for choosing one source over another is the need to match the isotope's maximum endpoint energy E_{max} , and consequently its linear attenuation coefficient μ , to the specific target thickness and density of the manufactured paper. As established in the theoretical analysis, the attenuation must be perfectly balanced: it must be sensitive to microscopic variations without being completely absorbed or fully transmitted.

Based on the simulation data, this engineering choice is implemented as follows:

- **For very thin, lightweight materials** (e.g., tissue paper or tracing paper), a lower-energy source like ^{90}Sr ($E_{max} = 546$ keV) is required. Because it has the highest attenuation coefficient ($\mu \approx 47.60$ cm $^{-1}$), its signal drops significantly even with minimal thickness variations, providing extremely high sensitivity. A higher-energy source would pass straight through these thin layers with almost zero attenuation, rendering the sensor "blind" to small defects (similar to the gamma-ray effect).
- **For standard printing paper**, ^{85}Kr provides an optimal middle ground ($E_{max} = 687$ keV, $\mu \approx 33.00$ cm $^{-1}$), offering a stable and measurable transmission profile.
- **For thicker paper or light cardboard**, a higher-energy source like ^{204}Tl ($E_{max} = 763$ keV) is fundamental. It possesses the lowest attenuation coefficient ($\mu \approx 30.00$ cm $^{-1}$) and the greatest penetrating power. If ^{90}Sr were used for cardboard, its low-energy electrons would be completely absorbed within the first few layers ($I(x) \rightarrow 0$), reducing the transmitted intensity to zero and completely blinding the sensor (similar to the lead absorber limitation).